A Complete Classification of Pythagorean Triples via Generalized Fibonacci Sequences and Pisano Periods

Will Dale

Independent Researcher

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Abstract

We show that the classical Barning–Berggren matrices generating the ternary tree of primitive Pythagorean triples arise as the quadratic lift of a semigroup of 2×2 integer matrices acting on a new coordinate system, the *BEDA parametrization* (b, e, d, a) with $d = b + e$, $a = b + 2e$. This structural explanation yields a complete classification of Pythagorean triples into five families—Fibonacci, Lucas, Phibonacci, Tribonacci, and Ninbonacci—corresponding to orbits of the Fibonacci matrix $F = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ acting on $(\mathbb{Z}/9\mathbb{Z})^2$; the modulus $9 = 3^2$ arises because 3 is inert in $\mathbb{Z}[\varphi]$, and a Hensel lift from mod-3 to mod-9 splits the single mod-3 cosmos orbit into three. The BEDA parametrization also exposes a complete *excircle algebra*: the four circle radii (r, r_C, r_F, r_G) form a rank-one 2×2 matrix satisfying $r \cdot r_G = r_C \cdot r_F = A$ (the triangle area) and a Lorentz-type identity $r_G^2 - r_F^2 - r_C^2 + r^2 = 4A$. As a corollary, every triangle with prescribed inradius $r = be$ is recovered by integer factorization of r alone. All claims are machine-verified (`verify_classification.py`, 10 checks, <1 second, stdlib only).

1 Introduction

1.1 Background on Pythagorean Triples

A *Pythagorean triple* is an ordered triple of positive integers (C, F, G) satisfying the Pythagorean theorem:

$$C^2 + F^2 = G^2 \tag{1}$$

The classical parametric representation of primitive Pythagorean triples, known since Euclid, is given by:

$$C = m^2 - n^2, \quad F = 2mn, \quad G = m^2 + n^2 \tag{2}$$

where $m > n > 0$, $\gcd(m, n) = 1$, and $m \not\equiv n \pmod 2$.

While this parametrization is complete, it does not reveal the deeper structural patterns that organize primitive triples. A complementary perspective arises from the ternary tree construction discovered by Berggren [6] and independently studied by Barning [5], in which

every primitive triple is generated exactly once via the action of three matrices in $\mathrm{GL}_3(\mathbb{Z})$ on the seed triple $(3, 4, 5)$. Despite its elegance, the algebraic origin of those generators has remained obscure.

In this paper we introduce the *BEDA parametrization*, a two-parameter coordinate system (b, e) for primitive triples, and show that the Barning–Berggren generators arise as the quadratic lift of a natural semigroup of 2×2 integer matrices acting on BEDA seeds. This structural explanation also reveals a five-fold family classification and a complete excircle algebra.

1.2 The BEDA Framework

The *BEDA tuple* is defined by four parameters (b, e, d, a) satisfying:

$$d = b + e \tag{3}$$

$$a = b + 2e \tag{4}$$

From any BEDA tuple, we generate a Pythagorean triple via:

$$C = 2de \tag{5}$$

$$F = ab \tag{6}$$

$$G = e^2 + d^2 \tag{7}$$

Proposition 1. *Every BEDA tuple (b, e, d, a) satisfying equations (3)–(4) generates a valid Pythagorean triple via equations (5)–(7).*

Proof. We verify $C^2 + F^2 = G^2$:

$$\begin{aligned}
C^2 + F^2 &= (2de)^2 + (ab)^2 \\
&= 4d^2e^2 + a^2b^2 \\
&= 4(b+e)^2e^2 + (b+2e)^2b^2 \\
&= 4(b^2e^2 + 2be^3 + e^4) + (b^4 + 4b^3e + 4b^2e^2) \\
&= 4b^2e^2 + 8be^3 + 4e^4 + b^4 + 4b^3e + 4b^2e^2 \\
&= b^4 + 4b^3e + 8b^2e^2 + 8be^3 + 4e^4
\end{aligned}$$

Meanwhile:

$$\begin{aligned}
G^2 &= (e^2 + d^2)^2 = (e^2 + (b+e)^2)^2 \\
&= (e^2 + b^2 + 2be + e^2)^2 = (b^2 + 2be + 2e^2)^2 \\
&= b^4 + 4b^3e + 8b^2e^2 + 8be^3 + 4e^4
\end{aligned}$$

Therefore $C^2 + F^2 = G^2$. □

1.3 Digital Roots and Pisano Periods

The *digital root* of a positive integer n, denoted $\mathrm{dr}(n)$, is defined as:

$$\mathrm{dr}(n) = \begin{cases} 9 & \text{if } n \equiv 0 \pmod 9 \\ n \bmod 9 & \text{otherwise} \end{cases} \tag{8}$$

The digital root maps integers to the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

For a Fibonacci-like sequence $\{F_n\}$ defined by the recurrence $F_n = F_{n-1} + F_{n-2}$ with initial conditions $F_0 = a, F_1 = b$, the *Pisano period* $\pi(m)$ is the period with which the sequence repeats when taken modulo m. In this work, we focus specifically on $m = 9$.

Contributions

The contributions of this paper are threefold.

1. **Structural explanation of the Barning–Berggren tree.** We prove an *Intertwining Theorem* (Theorem 5) showing that the classical 3×3 Barning–Berggren generators are the quadratic lift of a semigroup of 2×2 unimodular matrices acting on BEDA seeds. This gives the first structural explanation of why those matrices preserve the Pythagorean constraint.

2. **Complete orbit classification.** Using the BEDA coordinate system we classify all Pythagorean triples into five families (Theorems 3 and 4), identified as the orbits of the Fibonacci matrix on $(\mathbb{Z}/9\mathbb{Z})^2$. The modulus $9 = 3^2$ arises from an inert-prime Hensel lift in $\mathbb{Z}[\varphi]$, connecting the classification to the arithmetic of the golden-ratio field $\mathbb{Q}(\sqrt{5})$.

3. **Excircle algebra and inradius navigation.** The BEDA parametrization yields closed-form expressions for all four excircle radii as polynomial functions of (b, e, d, a) (Theorems 9–11). In particular, the inradius identity $r = be$ (Theorem 7) reduces the search for all triangles with a prescribed inradius to integer factorization, replacing Barning–Berggren tree search.

2 The Five Families

Definition 1 (The Five Families)**.** *We define five generalized Fibonacci sequences, each with initial terms (F_0, F_1) and recurrence relation $F_n = F_{n-1} + F_{n-2}$:*

1. ***Fibonacci Family**: $(F_0, F_1) = (1, 1)$*
2. ***Lucas Family**: $(F_0, F_1) = (2, 1)$*
3. ***Phibonacci Family**: $(F_0, F_1) = (3, 1)$*
4. ***Tribonacci Family**: $(F_0, F_1) = (3, 3)$*
5. ***Ninbonacci Family**: $(F_0, F_1) = (9, 9)$*

Each family generates a sequence of (b, e) pairs via $(b, e) = (F_n, F_{n+1})$.

2.1 Pisano Periods of the Five Families

Theorem 2 (Pisano Periods under Modulo 9)**.** *The five families exhibit the following Pisano periods when their digital roots are computed modulo 9:*

Family	***Initial Terms***	***Pisano Period*** $\pi(9)$
Fibonacci	$(1, 1)$	*24*
Lucas	$(2, 1)$	*24*
Phibonacci	$(3, 1)$	*24*
Tribonacci	$(3, 3)$	*8*
Ninbonacci	$(9, 9)$	*1*

Proof. The Pisano periods are established through direct computation. For each sequence, we compute the digital roots of consecutive terms and detect when the pattern first repeats. The computations are provided in Appendix A, with full verification available in the accompanying code repository.

For example, the Fibonacci sequence modulo 9 produces the digital root cycle:

$$\{1, 1, 2, 3, 5, 8, 4, 3, 7, 1, 8, 9, 8, 8, 7, 6, 4, 1, 5, 6, 2, 8, 1, 9\}$$

which repeats after 24 terms. □

3 Main Result: Complete Partition

Theorem 3 (Complete Partition of Digital Root Pairs)**.** *The five families partition the set of all possible digital root pairs* $\{1, 2, \ldots, 9\}^2$ *into five disjoint subsets. Specifically:*

1. *The Fibonacci family contributes 24 unique* $(dr(b), dr(e))$ *pairs*
2. *The Lucas family contributes 24 unique* $(dr(b), dr(e))$ *pairs*
3. *The Phibonacci family contributes 24 unique* $(dr(b), dr(e))$ *pairs*
4. *The Tribonacci family contributes 8 unique* $(dr(b), dr(e))$ *pairs*
5. *The Ninbonacci family contributes 1 unique* $(dr(b), dr(e))$ *pair*

The total is $24 + 24 + 24 + 8 + 1 = 81 = 9^2$*, accounting for all possible digital root pairs with* no overlaps.

Proof. We establish this result computationally by:

1. Computing the Pisano cycles for each of the five families
2. Extracting the unique $(\mathrm{dr}(F_n), \mathrm{dr}(F_{n+1}))$ pairs from each cycle
3. Verifying that the union of these five sets contains exactly 81 pairs
4. Verifying that the intersection of any two distinct sets is empty

The complete classification table is provided in Table 1. □

Table 1: Complete Classification of Digital Root Pairs by Family

dr(b)\dr(e)	1	2	3	4	5	6	7	8	9
1	F	F	L	P	F	P	L	F	F
2	L	L	F	L	P	P	L	F	L
3	P	P	T	L	F	T	F	L	T
4	F	P	F	P	P	L	L	P	P
5	P	L	L	P	P	F	P	F	P
6	L	F	T	F	L	T	P	P	T
7	F	L	P	P	L	F	L	L	L
8	F	L	P	F	P	L	F	F	F
9	F	L	T	P	P	T	L	F	N

F = Fibonacci, L = Lucas, P = Phibonacci, T = Tribonacci, N = Ninbonacci

4 Implications and Orbital Structure

4.1 The 72-8-1 Structure

The five families naturally group into three *orbital classes* based on their Pisano periods:

- **24-Cycle Orbit ("Cosmos")**: Fibonacci + Lucas + Phibonacci = $24+24+24 = 72$ pairs
- **8-Cycle Orbit ("Satellite")**: Tribonacci = 8 pairs
- **1-Cycle Orbit ("Singularity")**: Ninbonacci = 1 pair (the fixed point $(9, 9)$)

We now give a group-theoretic proof that this structure is not coincidental.

4.2 Group-Theoretic Origin: Five Families as Orbits of the Fibonacci Matrix

Let $\mathbb{Z}/9\mathbb{Z}$ denote the integers modulo 9, with elements $\{0, 1, \ldots, 8\}$ (where 0 corresponds to digital root 9). The digital root map $\mathrm{dr} : \mathbb{Z}_{>0} \to \{1, \ldots, 9\}$ induces a well-defined map $(\mathrm{dr}(b), \mathrm{dr}(e)) \in (\mathbb{Z}/9\mathbb{Z})^2$.

Define the *Fibonacci matrix*

$$F = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \in \mathrm{GL}_2(\mathbb{Z}/9\mathbb{Z}),$$

which acts on $(\mathbb{Z}/9\mathbb{Z})^2$ by $(b, e) \mapsto (e,\, b + e)$. This is exactly the Fibonacci recurrence step.

Theorem 4 (Group-Theoretic Classification)**.** *The five families of Theorem 3 are exactly the orbits of the cyclic group $\langle F \rangle$ acting on $(\mathbb{Z}/9\mathbb{Z})^2$. Specifically:*

1. *F has order 24 in $\mathrm{GL}_2(\mathbb{Z}/9\mathbb{Z})$, with $\det F = -1$ and $F^{24} = I$.*

2. *The action produces exactly* 5 *orbits with sizes* 24, 24, 24, 8, 1*, matching the five families.*

3. *The unique fixed point* $(0,0)$ *is the Ninbonacci family; the length-*8 *orbit is the Tribonacci family; the three length-*24 *orbits are the Fibonacci, Lucas, and Phibonacci families.*

4. *The three cosmos orbits are distinguished by the* $Q(\sqrt{5})$ norm $N(b,e) = b^2 + be - e^2$ *reduced mod* 9*: since* $\det F = -1$ *the action maps* $N \mapsto -N$*, so each cosmos orbit carries a norm pair* $\{N, -N\}$ *with values*

$$\textit{Fibonacci}: \{1,8\}, \quad \textit{Lucas}: \{4,5\}, \quad \textit{Phibonacci}: \{2,7\}.$$

5. *The Tribonacci orbit satisfies* $N(b,e) \equiv 0 \pmod 9$ *for all its elements; its stabilizer is* $\langle F^8 \rangle \cong \mathbb{Z}/3\mathbb{Z}$*.*

Proof. All claims are verified by exhaustive computation over the 81 elements of $(\mathbb{Z}/9\mathbb{Z})^2$, which we describe explicitly.

Order of F*.* Direct matrix exponentiation mod 9 gives $F^{24} = I$ and $F^k \neq I$ for $k < 24$; note $\det F = -1 \equiv 8 \pmod 9$, so $F \notin \mathrm{SL}_2(\mathbb{Z}/9\mathbb{Z})$ but $F^2 \in \mathrm{SL}_2(\mathbb{Z}/9\mathbb{Z})$.

Orbit computation. Starting from each $(b,e) \in (\mathbb{Z}/9\mathbb{Z})^2$ and iterating F until the orbit closes, one obtains exactly 5 distinct orbits with sizes $1 + 8 + 24 + 24 + 24 = 81$.

Matching to families. The initial conditions $(b_0, e_0) = (1,1), (2,1), (3,1), (3,3), (0,0)$ (in $\mathbb{Z}/9\mathbb{Z}$ coordinates) belong to distinct orbits. Cross-checking each orbit against the classification of Theorem 3 confirms that every element of a given orbit carries the same family label, and distinct orbits carry distinct labels.

Norm invariant. For the Fibonacci step $F : (b,e) \mapsto (e, b+e)$,

$$N(e, b+e) = e^2 + e(b+e) - (b+e)^2 = -(b^2 + be - e^2) = -N(b,e).$$

So F negates the norm. On a length-24 orbit $F^{24} = I$ multiplies the norm by $(-1)^{24} = 1$, consistent. The norm values $\{N, -N \pmod 9\}$ are thus constant on each orbit, and computation gives the three pairs stated. □

Remark 1 (Pisano period explained)**.** *The order of* F *in* $\mathrm{GL}_2(\mathbb{Z}/9\mathbb{Z})$ *equals the Pisano period* $\pi(9) = 24$*: both measure the period of the Fibonacci sequence modulo* 9*. Theorem 4 thus explains* why $\pi(9) = 24$ *is the "right" period for the five-family classification — it is the order of the Fibonacci generator acting on the digital-root state space. The 72-8-1 decomposition is the orbit-type decomposition of this cyclic group action.*

Remark 2 (GF(9) and the role of modulus 9)**.** *The choice of modulus 9 has a two-step arithmetic explanation.*

Step 1 (inert prime). *The rational prime* 3 *is* inert *in* $\mathbb{Z}[\varphi]$*: the minimal polynomial* $x^2 - x - 1$ *of* $\varphi = \frac{1+\sqrt{5}}{2}$ *is irreducible over* $\mathbb{F}_3$ *(discriminant* $5 \equiv 2 \pmod 3$ *is a non-square). Therefore* $\mathbb{Z}[\varphi]/3\mathbb{Z}[\varphi] \cong \mathrm{GF}(9)$*, and* F *represents* $\times\varphi$ *in* $\mathrm{GF}(9)$*. The order of* F *in* $\mathrm{GL}_2(\mathbb{F}_3)$ *equals* $\mathrm{ord}(\varphi) = 2(3+1) = 8$*. Consequently,* F *acting on* $(\mathbb{Z}/3\mathbb{Z})^2$ *has exactly two orbits: one cosmos orbit of size* 8 *and the fixed point* $(0,0)$*.*

Step 2 (Hensel lift to mod 9). *Lifting from mod-3 to mod-9 $= 3^2$ splits the single mod-3 cosmos orbit into three cosmos orbits of size 24, and introduces the Tribonacci orbit (size 8, corresponding to elements with $N \equiv 0 \pmod 9$). The five families are thus a mod-9 phenomenon, not a mod-3 phenomenon; modulus $9 = 3^2$ is the first level at which the full five-fold structure appears.*

The companion paper [7] extends this analysis to all primes p, covering the inert, split, and ramified cases.

Remark 3 (Geodesic–horocycle decomposition)**.** *The BEDA coordinate system makes a classical structure in the modular surface $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ simultaneously visible for both the five-family classification and the Barning–Berggren triple tree.*

The QA step $T = F^2 = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right]$ has trace 3 and determinant 1, so it is a hyperbolic *element of $\mathrm{SL}_2(\mathbb{Z})$: its two real fixed points are $1/\varphi$ and $-\varphi$ (the golden ratio and its conjugate on the real axis). The* geodesic flow *generated by T runs along the hyperbolic geodesic in $\mathbb{H}$ connecting $-\varphi$ to $1/\varphi$; modulo 9 its orbits on $(\mathbb{Z}/9\mathbb{Z})^2$ are exactly the five families of Theorem 4.*

The Berggren child operators $B = \left[\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right]$ and $C = \left[\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right]$ (in BEDA (b, e) coordinates) each have trace 2 and determinant 1, making them parabolic *elements of $\mathrm{SL}_2(\mathbb{Z})$: they fix the cusps 0 and ∞ respectively and generate the* horocycle flow*. The Barning–Berggren tree is the orbit of the seed $(b, e) = (1, 1)$ under these parabolic generators.*

These two flows are orthogonal foliations *of the modular surface:*

$$\underbrace{\langle T\rangle\text{-orbits}}_{\text{five families (geodesic)}} \quad \perp \quad \underbrace{\langle B, C\rangle\text{-orbits}}_{\text{triple tree (horocycle)}} .$$

The BEDA parametrization is the coordinate system in which both foliations are explicit and computable. This connection to the geodesic/horocycle structure of $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ suggests that the equidistribution results for Pythagorean triples (analogues of Weyl's theorem for horocycle flows) should have natural interpretations in terms of the five-family partition.

4.3 The Barning–Berggren Intertwining Theorem

The geodesic–horocycle decomposition of Remark 3 identifies three child operators acting on BEDA pairs $(b, e) \in \mathbb{Z}^2_{>0}$:

$$M_A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad M_B = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}, \quad M_C = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \tag{9}$$

where $M_X \cdot (b, e)^T$ produces the BEDA seed of the X-th Berggren child. Let $\tau : \mathbb{Z}^2_{>0} \to \mathbb{Z}^3_{>0}$ denote the *BEDA map*

$$\tau(b, e) = \big(2(b + e)e,\ (b + 2e)b,\ e^2 + (b + e)^2\big) = (C, F, G).$$

Theorem 5 (Barning–Berggren Intertwining)**.** *The classical Barning–Berggren 3×3 matrices*

$$R_A = \begin{pmatrix} -1 & 2 & 2 \\ -2 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix}, \quad R_B = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{pmatrix}, \quad R_C = \begin{pmatrix} 1 & -2 & 2 \\ 2 & -1 & 2 \\ 2 & -2 & 3 \end{pmatrix}$$

satisfy the intertwining relation

$$\tau(M_X \cdot u) = R_X \cdot \tau(u) \qquad \text{for all } u = (b,e)^T \in \mathbb{Z}^2_{>0}, \quad X \in \{A,B,C\}.$$

Consequently τ is a semigroup morphism: for every composition $W = M_{X_1} \cdots M_{X_k}$ and every seed u,

$$\tau(W \cdot u) = R_{X_1} \cdots R_{X_k} \cdot \tau(u).$$

The Barning–Berggren Pythagorean triple tree is therefore the quadratic lift *of the* $\mathrm{SL}_2(\mathbb{Z})$ *semigroup $\langle M_A, M_B, M_C\rangle$ acting on BEDA seeds.*

Proof. For each generator the identity $\tau(M_X \cdot u) = R_X \cdot \tau(u)$ is a polynomial identity in $(b,e) \in \mathbb{Z}^2$. We verify it for M_A; the other two cases are analogous.

Under $M_A : (b,e) \mapsto (b,\, b+e)$, set $b' = b$, $e' = b+e = d$. Then $d' = b'+e' = b+d = 2b+e$ and the new triple is

$$C' = 2d'e' = 2(2b+e)(b+e), \quad F' = a'b' = (3b+2e)b, \quad G' = {e'}^2 + {d'}^2 = d^2 + a^2.$$

Expanding $R_A \cdot (C,F,G) = (-C+2F+2G,\, -2C+F+2G,\, -2C+2F+3G)$ via $C = 2de$, $F = ab$, $G = e^2+d^2$ and simplifying using $d = b+e$, $a = b+2e$ yields identical expressions for each component, confirming $\tau(M_A u) = R_A \tau(u)$. The composition claim follows by induction.

All three generator identities were also verified computationally over 25 independent seeds each. Starting from the seed $(b,e) = (1,1)$, the full Barning–Berggren tree under $\langle M_A, M_B, M_C\rangle$ generates all 80 primitive Pythagorean triples with $G \le 500$, with 0 missed. □

Remark 4. *Theorem 5 gives a structural explanation for why the 3×3 Barning–Berggren matrices preserve $C^2+F^2-G^2 = 0$: the Pythagorean constraint is the degree-2 relation $\tau^*(C^2+F^2-G^2) = 0$, which is preserved by τ by construction, and the 3×3 action is merely the induced action on the image. In particular, the BEDA parametrization supplies the "missing" quadratic model that makes the Barning–Berggren tree algebraically transparent.*

4.4 Connection to Pythagorean Triples

Corollary 6. *Every Pythagorean triple (C,F,G) generated from a BEDA tuple (b,e,d,a) can be uniquely classified into one of the five families based on $(dr(b), dr(e))$.*

This provides a new perspective on the structure of Pythagorean triples, complementary to the classical (m,n) parametrization.

4.5 Inradius as a Navigation Coordinate

The BEDA parametrization exposes a striking invariant: the inradius of every generated right triangle equals the product be.

Theorem 7 (Inradius Identity)**.** *For any BEDA tuple (b,e,d,a) satisfying* (3)–(4)*, the associated Pythagorean right triangle (C,F,G) has inradius*

$$r = b \cdot e.$$

Proof. The inradius of a right triangle with legs C, F and hypotenuse G is

$$r = \frac{C + F - G}{2}.$$

Expanding each side using $d = b + e$ and $a = b + 2e$:

$$\begin{aligned} C &= 2de = 2e(b + e) = 2be + 2e^2, \\ F &= ab = (b + 2e)b = b^2 + 2be, \\ G &= e^2 + d^2 = e^2 + (b + e)^2 = b^2 + 2be + 2e^2. \end{aligned}$$

Therefore

$$C + F - G = (2be + 2e^2) + (b^2 + 2be) - (b^2 + 2be + 2e^2) = 2be,$$

and $r = 2be/2 = be$. □

Corollary 8 (Inradius Navigation)**.** *Given a positive integer r, the complete set of BEDA-generated Pythagorean right triangles with inradius r is in bijection with the ordered factor pairs (b, e) of r with $b, e > 0$. For each such pair, the full tuple and triple are recovered deterministically as*

$$d = b + e, \quad a = b + 2e, \quad C = 2de, \quad F = ab, \quad G = e^2 + d^2.$$

Proof. Theorem 7 shows $r = be$ is necessary. Conversely, every positive factorization $r = be$ produces a valid BEDA tuple (both d and a are positive integers), which by Proposition 1 generates a valid Pythagorean triple. The map $(b, e) \mapsto (C, F, G)$ is injective since distinct ordered pairs (b, e) yield distinct pairs $(b^2+2be, 2be+2e^2)$ for F and C respectively. □

Remark 5 (Connection to the Barning–Berggren Tree)**.** *The Barning–Berggren ternary tree [5, 6] generates every primitive Pythagorean triple exactly once via three unimodular* 3×3 *matrices acting on (C, F, G). Theorem 7 shows that this tree is simultaneously traversing the* divisor lattice *of $r = be$: each tree node corresponds to a pair (b, e), and the three child transforms correspond to divisor-lattice moves. The BEDA coordinate system thus provides a flat, arithmetic parameterization of what the Barning–Berggren tree explores hierarchically. In particular, a tree search for triples with a given inradius r reduces to integer factorization of r—a substantial algorithmic simplification.*

Example 1. *Take $r = 6$. The factor pairs are $(1, 6), (2, 3), (3, 2), (6, 1)$, giving four BEDA seeds:*

(b, e)	d	a	C	F	G
$(1, 6)$	7	13	84	13	85
$(2, 3)$	5	8	30	16	34
$(3, 2)$	5	7	20	21	29
$(6, 1)$	7	8	14	48	50

Each triple is verified to satisfy $C^2 + F^2 = G^2$ and to have inradius 6.

5 Complete Excircle Algebra

The Inradius Identity $r = be$ of Theorem 7 is the first member of a complete excircle system encoded by every BEDA tuple. We derive the full system, establish a semiperimeter formula, and prove a quadratic identity among the four radii.

5.1 The Four Radii

Recall that for a triangle with area A, semiperimeter s, and sides C, F, G, the three exradii satisfy

$$r_C = \frac{A}{s-C}, \quad r_F = \frac{A}{s-F}, \quad r_G = \frac{A}{s-G}.$$

Theorem 9 (Excircle Identities)**.** *For any BEDA tuple (b, e, d, a) with $d = b+e$, $a = b+2e$, the associated right triangle $(C, F, G) = (2de,\ ab,\ e^2 + d^2)$ satisfies:*

1. $s = da$ (semiperimeter equals the outer product),
2. $r_C = ae, \quad r_F = bd, \quad r_G = ad = s,$
3. $A = abde$ (area is the product of all four tuple entries).

Proof. Semiperimeter. Using $d = b + e$, $a = b + 2e$:

$$\begin{aligned} 2s = C + F + G &= 2de + ab + e^2 + d^2 \\ &= 2e(b+e) + b(b+2e) + e^2 + (b+e)^2 \\ &= 2b^2 + 6be + 4e^2 = 2(b+e)(b+2e) = 2da, \end{aligned}$$

so $s = da$.

Area. $A = CF/2 = (2de)(ab)/2 = abde$. The incircle formula checks out: $rs = be \cdot da = abde = A$.

Exradii. From $A = r_X(s - X)$:

$$\begin{aligned} &s - C = da - 2de = d(a - 2e) = db, \quad \Rightarrow \quad r_C = \tfrac{abde}{db} = ae. \\ &s - F = da - ab = a(d - b) = ae, \quad \Rightarrow \quad r_F = \tfrac{abde}{ae} = bd. \\ &s - G = be \quad \text{(Theorem 7)}, \quad \Rightarrow \quad r_G = \tfrac{abde}{be} = ad = s. \end{aligned}$$

□

Remark 6. *The identity $r_G = s$ (exradius opposite the hypotenuse equals the semiperimeter) holds for* every *right triangle: since $r = (C + F - G)/2$ one has $s - G = r$, giving $r_G = A/r = rs/r = s$. The BEDA content is that $s = da$ and $r_G = ad$ supply explicit polynomial formulas in the tuple coordinates.*

5.2 The Rank-One Radius Structure

Theorem 10 (Rank-One Radius Matrix)**.** *The four radii of Theorem 9 form a rank-one 2×2 matrix:*

$$\begin{pmatrix} r & r_C \\ r_F & r_G \end{pmatrix} = \begin{pmatrix} be & ae \\ bd & ad \end{pmatrix} = \begin{pmatrix} b \\ a \end{pmatrix} \begin{pmatrix} e & d \end{pmatrix}.$$

In particular, $r \cdot r_G = r_C \cdot r_F = A$.

Proof. The outer-product factorization is immediate. The determinant of a rank-one matrix is zero:

$$r \cdot r_G - r_C \cdot r_F = be \cdot ad - ae \cdot bd = 0,$$

so $r \cdot r_G = r_C \cdot r_F = abde = A$. □

Remark 7. *Theorem 10 says the excircle system has only two independent degrees of freedom: the outer elements (b, a) and the inner elements (e, d) of the BEDA tuple. All four radii are recoverable from any one of them and the area $A = abde$.*

5.3 A Quadratic Identity among the Radii

Theorem 11 (Lorentz-Type Identity)**.**

$$r_G^2 - r_F^2 - r_C^2 + r^2 \;=\; 2CF \;=\; 4A.$$

Proof. Factor by grouping:

$$r_G^2 - r_F^2 - r_C^2 + r^2 = d^2(a^2 - b^2) - e^2(a^2 - b^2) = (a^2 - b^2)(d^2 - e^2).$$

Expand using the BEDA recurrences $a + b = 2d$, $a - b = 2e$, $d + e = a$, $d - e = b$:

$$a^2 - b^2 = (a + b)(a - b) = 2d \cdot 2e = 4de = 2C,$$

$$d^2 - e^2 = (d + e)(d - e) = a \cdot b = F.$$

Therefore $(a^2 - b^2)(d^2 - e^2) = 2C \cdot F = 4abde = 4A$. □

Remark 8 (Connection to Toroidal Geometry)**.** *The pair $(r,\, r_G) = (be,\, da)$ provides natural minor and major radii for a toroid associated to a BEDA triangle. Their product $r \cdot r_G = A$ (the triangle area) is the bipolar scale invariant $a^2 = Rr_{\text{minor}}$ of Volk's bipolar coordinate theory [8], and identifies the Appollonian circle family determined by the triangle with bipolar scale $\sqrt{A} = \sqrt{abde}$.*

Under the Fibonacci generator F, the five orbital families have periods $24, 8, 1$ (Theorem 4). In Volk's framework, stable toroidal structures correspond to torus knots $T(m, n)$ with integer winding numbers; transitions between knot types are quantum jumps. The orbit periods suggest that the three cosmos families (24-cycle), the Tribonacci satellite (8-cycle), and the Ninbonacci fixed point (1-cycle) correspond to three qualitatively distinct toroidal winding classes. A complete topological classification of these classes via the excircle coordinates (r, r_G, A) is left to future work.

Example 2. *Take $(b, e) = (3, 2)$, giving $d = 5$, $a = 7$ and triple $(C, F, G) = (20, 21, 29)$. The four radii are:*

$$r = 6, \quad r_C = 14, \quad r_F = 15, \quad r_G = 35 = s.$$

The area is $A = 210$, verified by $r_G^2 - r_F^2 - r_C^2 + r^2 = 1225 - 225 - 196 + 36 = 840 = 4 \times 210$.

6 Examples

We provide examples of Pythagorean triples from each family:

6.1 Fibonacci Family

From $(b, e) = (1, 1)$: $(C, F, G) = (4, 3, 5)$
From $(b, e) = (2, 3)$: $(C, F, G) = (30, 16, 34)$
From $(b, e) = (5, 8)$: $(C, F, G) = (208, 105, 233)$

6.2 Lucas Family

From $(b, e) = (2, 1)$: $(C, F, G) = (6, 8, 10)$
From $(b, e) = (3, 4)$: $(C, F, G) = (56, 33, 65)$
From $(b, e) = (7, 11)$: $(C, F, G) = (396, 203, 445)$

6.3 Tribonacci Family (8-Cycle)

From $(b, e) = (3, 3)$: $(C, F, G) = (36, 27, 45)$
From $(b, e) = (6, 9)$: $(C, F, G) = (270, 144, 306)$
From $(b, e) = (15, 24)$: $(C, F, G) = (1872, 945, 2097)$

6.4 Phibonacci Family

From $(b, e) = (3, 1)$: $(C, F, G) = (8, 15, 17)$
From $(b, e) = (4, 5)$: $(C, F, G) = (90, 56, 106)$
From $(b, e) = (9, 14)$: $(C, F, G) = (644, 333, 725)$

6.5 Ninbonacci Family (Fixed Point)

From $(b, e) = (9, 9)$: $(C, F, G) = (324, 243, 405)$
All multiples: $(648, 486, 810)$, $(972, 729, 1215)$, etc.

7 Applications

7.1 Computational Number Theory

The classification provides an efficient algorithm for generating and categorizing Pythagorean triples:

- Given $(\mathrm{dr}(b), \mathrm{dr}(e))$, Table 1 gives the family in $O(1)$.
- Given an inradius r, Corollary 8 yields all BEDA triangles with inradius r via integer factorization of r alone, replacing tree search.
- The norm invariant $\{N(b, e) \bmod 9\}$ distinguishes the three cosmos families, providing a fast membership test.

8 Conclusion and Future Work

We have established a complete classification of Pythagorean triples into five families based on generalized Fibonacci sequences and their Pisano periods under modulo 9 arithmetic. This classification reveals a 72-8-1 orbital structure with applications in number theory, cryptography, and dynamical systems.

Future research directions include:

- Extending the group-theoretic classification to other moduli: the three prime cases (inert $p \equiv \pm 2 \pmod 5$, split $p \equiv \pm 1 \pmod 5$, ramified $p = 5$) yield distinct orbit structures in $(\mathbb{Z}/p^k\mathbb{Z})^2$; the present paper is the inert case $p = 3$, $k = 2$. A companion paper [7] treats all three cases.
- Characterizing which factor pairs (b, e) of a given inradius r lie in each family: by Theorems 7 and 3 this reduces to classifying divisors of r by their digital-root pair.
- Making precise the equidistribution statement of Remark 3: by Ratner's theorem the horocycle flow on $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$ equidistributes, implying that primitive Pythagorean triples are asymptotically uniform across the three cosmos families (each receiving density 24/81 of all triples ordered by hypotenuse).
- Investigating connections to the theory of modular symbols and period integrals for the modular surface $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$.

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A Computational Verification

All results in this paper have been computationally verified using Python 3.x with NumPy. The complete source code, including:

- Generators for all five families
- Pisano period computation
- Partition verification
- Pythagorean triple generation and validation

is available at the accompanying repository. The standalone script `verify_classification.py` runs in under one second with no dependencies beyond the Python standard library. It checks all ten computational claims (V1–V10): partition completeness, Pisano periods, orbit–family correspondence, order of F, norm alternation, cosmos norm pairs, Tribonacci stabilizer, Ninbonacci fixed point, inradius identity, and generator actions.